

Team: B16

Domain: Machine Learning

Title: High-Performance Stroke Prediction Using Categorical Boosting And Explainable Machine Learning Models

ABSTRACT:

Stroke is a leading cause of mortality and long-term disability worldwide, posing serious health and economic challenges. With an aging global population, the incidence of stroke is steadily increasing, highlighting the need for accurate and timely prediction systems to support early intervention and improved patient outcomes.

This project presents an automated stroke prediction framework using machine learning algorithms, designed to enhance the precision of early diagnosis. The performance of the proposed model is compared against six widely used classifiers to evaluate accuracy, robustness, and generalization capability in predicting stroke risk.

To ensure model transparency, SHAP (Shapley Additive Explanations) is utilized, offering detailed insights into feature importance and model decision logic. This explainable AI approach is crucial for building trust and accountability in healthcare applications, where understanding the reasoning behind predictions is essential.

The study finds that complex models, when paired with interpretability techniques, outperform simpler methods while still maintaining explainability. A web application is also developed as part of the project, enabling real-time stroke prediction and demonstrating practical deployment in clinical or preventive settings.

Overall, this work contributes to the development of reliable, interpretable, and accessible tools for stroke risk assessment, supporting early diagnosis and facilitating data-driven healthcare decisions